

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	January 26, 2023
			Page	Page 1 of 20

(1) Research Program/Project/Study Title: Performance of Different Varieties of Sweet Potato (<i>Ipomoea batatas</i>) as Affected by CAPSU Formulated Soil Ameliorants	
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Program or Project has component (if no proceed to 2)? ☐ Yes ☐ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☒ Natural/Biological Sciences ☐ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster		(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☒ CSRDC Crop Science Research Development Center	
(4) Site of implementation			
Municipality/City:		Mambusao	
Province:		Capiz	
Region:		VI	
(5) Classification Research: ☐ Basic ☒ Applied Development: ☐ Pilot Testing ☐ Tech. Promotion/Commercialization		(6) Mode of Implementation ☒ Single Authorship/Proponent ☐ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)	
(7) Sector/Commodity Sweet Potato		(8) Discipline/Field of Specialization Agriculture	

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 2 of 20

(9A) Abstract (This has a maximum of 250 words. It contains 1-2 sentences of rationale, followed by objectives, brief methodology, major findings, and implications of impact of the study.)

The study was conducted at the Organic Agriculture project of Capiz State University - Burias Campus, Burias, Mambusao, Capiz. It was designed as a two-factor experiment laid out in a Randomized Complete Block Design (RCBD) with three replications. Factor A consisted of different sweet potato varieties: A1 - Maita, A2 - PSB SP 25, A3 - PSB SP 30, and A4 - PSB SP 13. Factor B comprised CAPSU Formulated Soil Ameliorants: B1 - CAPSU Formulated Soil Ameliorants (CAFSA) + Partially Decomposed Rice Hull (PDRH), B2 - CAFSA + Carbonized Rice Hull (CRH), and B3 - pure CAFSA. All data gathered were subjected to Analysis of Variance (ANOVA) using STAR software. Differences among treatments were determined using the Duncan Multiple Range Test (DMRT).

The results revealed that the growth and yield of different sweet potato varieties varied significantly when applied with CAFSA. Sweet potato varieties had a highly significant impact on the growth and yield parameters, except for stem girth. They also had a significant impact on the diameter of tubers and the number of nodes. The application of CAFSA had a significant effect on the weight of tubers but not on other growth and yield parameters. The application of CAFSA showed comparable results across treatments. There was no interaction effect between the different sweet potato varieties and the application of CAFSA.

(9B) Keywords (Write 3-6 words you used in the study, preferably not found in the title because words in the title are already given keywords. This would also include process, or methodology employed.)

organic farming, organic, soil ameliorants, CAFSA, sweet potato

- 1. Introduction** (This includes the rationale, background of the study, and objectives which are not divided by subheadings. Incorporate at least three (3) reviews)

Sweet potato (*Ipomoea batatas*) is a valuable and resilient crop that plays a vital role in global food security, particularly in regions with diverse agroecological conditions. The performance and productivity of sweet potatoes are influenced by various factors, including soil quality and fertility. Soil ameliorants have long been recognized as essential tools for improving soil conditions and enhancing crop yield (Fageria et al., 2018).

The selection of suitable soil ameliorants is a critical decision for agricultural practices as they can influence various soil properties, such as nutrient content and pH levels, which, in turn, can impact the growth, yield, and quality of sweet potato (Makinde et al., 2017). However, in recent years, Capiz State University (CAPSU) has formulated soil ameliorants, including bokashi, to improve soil conditions and support sustainable crop production. Bokashi, a form of organic soil ameliorant, has gained recognition for its potential to enhance soil fertility and nutrient availability (Ong et al., 2018). Bokashi is a fermented mixture of organic matter that contains beneficial microorganisms. When applied to the soil, it can improve soil structure, increase nutrient content, and foster a favorable environment for plant growth. Bokashi has shown promise in improving crop productivity by enhancing nutrient uptake, root development, and overall plant health.

The performance of different sweet potato varieties is known to vary depending on soil conditions and nutrient availability. Investigating how sweet potato varieties respond to the application of CAPSU-formulated soil ameliorants (CAFSA) is crucial for optimizing cultivation practices and improving crop productivity. Bokashi's impact on nutrient uptake, growth, and overall performance of sweet potato varieties is of particular interest, as it may contribute to enhanced yields and improved tuber quality.

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 3 of 20

This study aimed to assess the performance of different sweet potato varieties when subjected to the application of CAPSU formulated soil ameliorants (CAFSA). By examining how the CAPSU formulated soil ameliorants (CAFSA) affects the growth, yield, and quality of various sweet potato varieties, this research contributes to a better understanding of the potential benefits of using CAPSU formulated soil ameliorants (CAFSA), ultimately benefiting both producers and consumers.

Objectives of the Study

The study was conducted to determine the performance of different varieties of sweet potato as affected by CAPSU Formulated Soil Ameliorants (CAFSA). Specifically, the study aims to:

1. Determine the growth and yield performance of different sweet potato varieties applied with CAPSU Formulated Soil Ameliorants (CAFSA).
2. Evaluate the substrate that best promotes the growth and yield of different sweet potato varieties.
3. Determine the presence of interaction effects between the different sweet potato varieties and CAFSA.

(10) **Materials and Methods or Methodology** (This includes the materials, methods, duration & location of the study. Materials and Methods are headings used in Natural Science researches and methodology is Social Sciences. Presentation of materials may not necessarily be in separate heading but may be incorporated in the specific methods used. Justify paragraphs with main headings in caps.)

The experiment was carried out at the Organic Agriculture project of Capiz State University - Burias Campus, Burias, Mambusao, Capiz. It was designed as a two-factor experiment laid out in Randomized Complete Block Design (RCBD) with three replications. Factor A consisted of different sweet potato varieties such as; A1 - Maita, A2 - PSB SP 25, A3 - PSB SP 30, and A4 - PSB SP 13. On the other hand, Factor B comprised of CAPSU Formulated Soil Ameliorants (CAFSA): B1 - CAPSU Formulated Soil Ameliorants (CAFSA) + Partially Decomposed Rice Hull (PDRH), B2 - CAFSA + Carbonized Rice Hull (CRH), B3 - pure CAFSA. All data gathered were subjected to the Analysis of Variance (ANOVA) using STAR software. Differences among treatments was determined using the Duncan Multiple Range Test (DMRT).

(11) **Results and Discussions** (This part contains presentation of results (data) with reference to the table and figure, to be followed by discussion, with support of reviewed literature. Sequence shall be based on the objectives of the study. Tables are formatted without the vertical lines. Horizontal lines are set at the top and bottom portions as well as below table headings and above major subdivisions. Justify paragraphs with main headings in caps.)

Length of Stem

The ANOVA for the length of stems showed a highly significant difference among the different sweet potato varieties. However, there was no significant effect observed from the application of CAFSA.

Among the different sweet potato varieties, PSB SP 13 had the longest stem length, with a mean of 546.46 cm, which was comparable to PSB SP 25, with a mean of 505.94 cm. This was followed by PSB SP 30, with a mean of 395.85 cm, while the Maita variety had the shortest stem length of 268 cm. The mean length of the stem ranged from 408.98 cm to 466.76 cm for the application of CAFSA.

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 4 of 20

There was no interaction effect between the different varieties of sweet potatoes and the application of CAFSA.

Stem Girth

The Analysis of Variance indicated that the stem girth was not significantly affected by the various sweet potato varieties and the application of CAFSA. The means for the sweet potato varieties ranged from 4.03 mm to 5.66 mm, while for the application of CAFSA, it ranged from 4.50 mm to 5.48 mm.

Moreover, there was no interaction effect between the different sweet potato varieties and the application of CAFSA in affecting the stem girth.

Number of Nodes

The ANOVA for the number of nodes revealed a significant effect on the different sweet potato varieties, but no significant effect from the application of CAFSA was found.

In terms of the application of CAFSA, the means ranged from 70.91 to 110.34. Among the different sweet potato varieties, PSBN SP 13 had the highest number of nodes with a mean of 176.03. Maita (55.80) had the lowest number of nodes, which was comparable with PSB SP 30 (63.99) and PSB SP 25 (84.10).

Moreover, the analysis revealed that the two factors did not interact to influence the number of nodes of the different sweet potato varieties.

Length of the Longest Leaf

The analysis of variance showed that the length of the longest leaf was highly significantly affected by different sweet potato varieties but not by the application of CAFSA.

Among the sweet potato varieties, PSB SP 30 had the longest leaf with a mean of 11.66 cm, which was comparable to the Maita variety (11.20 cm) and the PSBN SP 13 variety (9.78 cm). The shortest leaf length was obtained by the PSB SP 25 variety (8.51 cm), which was also comparable to the PSBN SP 13 variety (9.78 cm). In terms of the application of CAFSA, the means ranged from 9.92 cm to 10.82 cm.

However, there was no significant interaction effect between different sweet potato varieties and the application of CAFSA on the length of the longest leaf.

Width of the Longest Leaf

The ANOVA of the width of the longest leaf revealed a highly significant effect of different sweet potato varieties. However, there was no significant effect from the application of CAFSA.

Among the different sweet potato varieties, Maita had the widest leaf with a mean of 13.42 cm, followed by PSB SP 30 (12.11 cm). The smallest leaf was obtained by PSB SP 25 (9.37 cm), which was comparable to PSBN SP 13 (9.40 cm). The application of CAFSA means ranged from 10.96 cm to 11.15 cm.

There was no interaction effect between the different varieties of sweet potato and the application of CAFSA.

Biomass

The analysis of variance for biomass revealed a highly significant effect on the different sweet potato varieties but not on the application of CAFSA.

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 5 of 20

In terms of the application of CAFSA, the means ranged from 590.26 g to 640.41 g. For the different sweet potato varieties, PSB SP 30 had the heaviest biomass with a mean of 729.45 g, which was comparable to the Maita variety (726.05 g). PSB SP 25 produced the lightest biomass with a mean of 469.50 g, which was comparable to PSBN SP 13 (557.53 g).

However, there was no interaction effect between the different varieties of sweet potato and the application of CAFSA.

Number of tubers

The analysis of variance for the number of tubers revealed a highly significant effect on the different sweet potato varieties but not on the application of CAFSA.

In terms of the application of CAFSA, the means ranged from 590.26 g to 640.41 g. For the different sweet potato varieties, the Maita variety (2.22) had the most number of tubers, followed by PSB SP 30 (1.59). PSB SP 13 produced the least number of tubers (0.59), which was comparable to PSB SP 25 (1.06). The number of tubers produced per variety was limited due to lack of water during that period, rendering the soil hard and deterring the water needed by the plants for growth and development.

There was no interaction effect between the different varieties of sweet potato and the application of CAFSA.

Diameter of tubers

The results of the ANOVA indicated a significant effect of sweet potato variety on the diameter of tubers, but not on the application of CAFSA.

Among the sweet potato varieties, Maita had the largest diameter with a mean of 50.66 cm, which was comparable to PSB SP 30 (48.71 cm). The smallest diameter was observed in PSB SP 30 with a mean of 20.11 cm, comparable to PSB SP 25 (25.93 cm). Regarding the application of CAFSA, the means ranged from 31.36 cm to 39.15 cm.

There was no interaction effect between the different varieties of sweet potato and the application of CAFSA.

Weight of tubers

The ANOVA revealed that the weight of tubers was highly significantly affected by the different sweet potato varieties and a significant effect from the application of CAFSA.

The sweet potato varieties applied with only CAFSA produced the heaviest tubers, with an average weight of 130.56 g, which was comparable to the sweet potato varieties applied with CFSA+ PDRH (101.78 g).

On the other hand, the sweet potato varieties applied with CAFSA and CRH produced the lightest tubers, with an average weight of 90.45 g, which was also comparable to the sweet potato varieties applied with CAFSA + PDRH (101.78 g).

For the different sweet potato varieties, the Maita variety produced the heaviest tuber with an average weight of 190.70 g, followed by PSB SP 30 (137.69 g). PSB SP 13 produced the lightest tuber with an average weight of 48.58 g, which was comparable to PSB SP 25 (53.38 g). ANOVA further indicated that there was no significant interaction effect between the different sweet potato varieties and the application of CAFSA.

Hence, the two factors did not interact in influencing the weight of tubers of the different sweet potato varieties.

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 6 of 20

Length of Tubers

The analysis of variance for the length of tubers revealed a highly significant effect of the different sweet potato varieties and no significant effect from the application of CAFSA.

In terms of the application of CAFSA, the means ranged from 10.44 cm to 13.17 cm. Among the different sweet potato varieties, Maita produced the longest tubers (20.19 cm), followed by PSB SP 30 (14.51 cm). PSBN SP 13 produced the smallest tuber (4.23 cm), which was comparable to tubers produced by PSB SP 25 (7.31 cm).

However, there was no interaction effect between the two factors.

Number of Marketable tuber

The ANOVA for the number of marketable tubers revealed a highly significant effect on the different varieties of sweet potato but no significant effect from the application of CAFSA.

Among the different sweet potato varieties, Maita produced the greatest number of marketable tubers with a mean of 1.70, which was comparable to PSB SP 30 (1.24), which was likewise statistically comparable to PSB SP 25 (0.83). The lowest number of marketable tubers was produced by PSBN SP 13, which was also comparable to PSB SP 25 (0.83). On the other hand, the application of CAFSA means ranged from 1.02 to 1.13.

There was no interaction effect between the different sweet potato varieties and the application of CAFSA.

Number of Non-Marketable tuber

The ANOVA for the number of non-marketable tubers revealed a highly significant effect on the different varieties of sweet potato but no significant effect from the application of CAFSA.

In terms of the CAFSA, the means ranged from 0.52 to 0.74. For the different sweet potato varieties, Maita produced the greatest number of non-marketable tubers with a mean of 0.99, followed by PSB SP 25 (0.84) and PSB SP 30 (0.42). The lowest number of non-marketable tubers was produced by PSBN SP 13 with a mean of 0.22.

There was no interaction effect between the different sweet potato varieties and the application of CAFSA.

Yield per Square Meter

The analysis of variance for the yield per square meter revealed a highly significant effect on the different varieties of sweet potato but no significant effect from the application of CAFSA.

Among the different sweet potato varieties, the Maita variety produced the highest yield per square meter (1552.76 g), followed by PSB SP 30 (716.11 g), which was comparable to PSB SP 25 (650.56 g). The lowest yield per square meter was observed for PSB SP 13 (216.11 g), which was comparable to PSB SP 25 (650.56 g). In terms of the application of CFSA, the means ranged from 750 g to 802.09 g.

There was no interaction effect between the two factors.

(13) Conclusions and Recommendations (This part shall be written in narrative form and arranged based on the objectives and findings. Conclusions are general statements answering the objectives. Recommendations are based on the conclusions. They may be split or combined based on their length, which means, conclusions and recommendations may be split under separate headings if they are long. Justify paragraphs with main headings in caps.)

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 7 of 20

Conclusions

Based on the results of the study, the following conclusion were drawn:

1. The growth and yield of different sweet potato varieties varied significantly when applied with CAFSA.
2. Sweet potato varieties had a highly significant impact on the growth and yield parameters, especially on the diameter of tubers and the number of nodes.
3. Among the varieties of sweet potatoes tested, Maita gave the best results in almost all growth and yield parameters.
4. The application of CAFSA had a significant effect on the weight of tubers but not on other growth and yield parameters.
5. There was no interaction effect between the different sweet potato varieties and the application of CAFSA.

Recommendations

Based on the results of this study, the following recommendations are hereby forwarded:

1. Grow Maita and PSB SP 30 sweet potato varieties for higher yield when using CAFSA.
2. Use any substrate containing CAFSA in growing sweet potatoes.
3. Analyze the physical, chemical, and microbial properties of soil before and after cropping.
4. Conduct similar studies in an open field, supply abundant water, and during rainy season.
5. Validate and conduct similar studies on other vegetables and root crops.

(14) Output (Indicate the specific products, processes or services which the project has produced; quantify when possible.)

(15) Technology Transfer Activity Conducted (Indicate the specific products, activity design, processes or services transferred as extension activities)

(16) Economic Analysis (This should present the direct cost of project implementation, expenses in the production of technology, revenues generated from technology produced, return on investment (ROI) and payback period.)

(17) Acknowledgments (Include only important persons and institutions with involvement in funding and conduct of the research. Justify paragraphs with main headings in caps.)

(18) References (It should include only works mentioned in the text and should be arranged alphabetically. Citations shall not be less than 13 entries. This must follow the APA latest edition format, in hanging indention with 12-point space in between.)

References

- Ong, Y. T., Tan, C. S., & Leong, Y. K. (2018). Effect of bokashi fertilizer on the growth, yield, and postharvest quality of okra (*Abelmoschus esculentus*). *The Scientific World Journal*, 2018, 1-8.
- Makinde, A. A., Ayuba, K., Bada, B. S., & Bello, F. (2017). Nutrient composition and acceptability of 'amala' produced from high yield varieties of yam, cocoyam, cassava and sweet potato. *Food Science and Quality Management*, 65, 24-28.
- Ong, Y. T., Tan, C. S., & Leong, Y. K. (2018). Effect of bokashi fertilizer on the growth, yield, and postharvest quality of okra (*Abelmoschus esculentus*). *The Scientific World Journal*, 2018, 1-8.

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	January 26, 2023
			Page	Page 8 of 20

APPENDICES

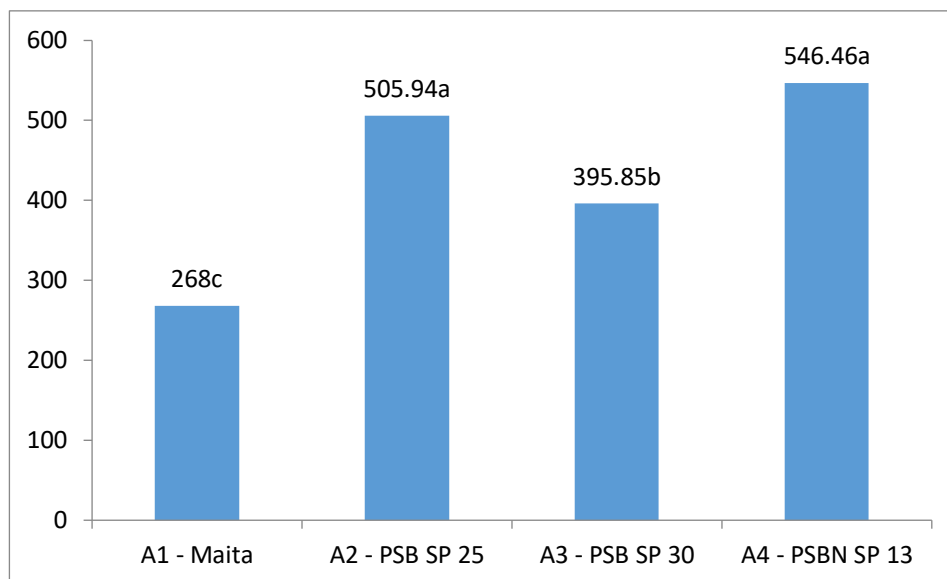


Figure 1. Length of stem of sweet potato varieties as affected by the different CAPSU Formulated Soil Ameliorants (CFSA).

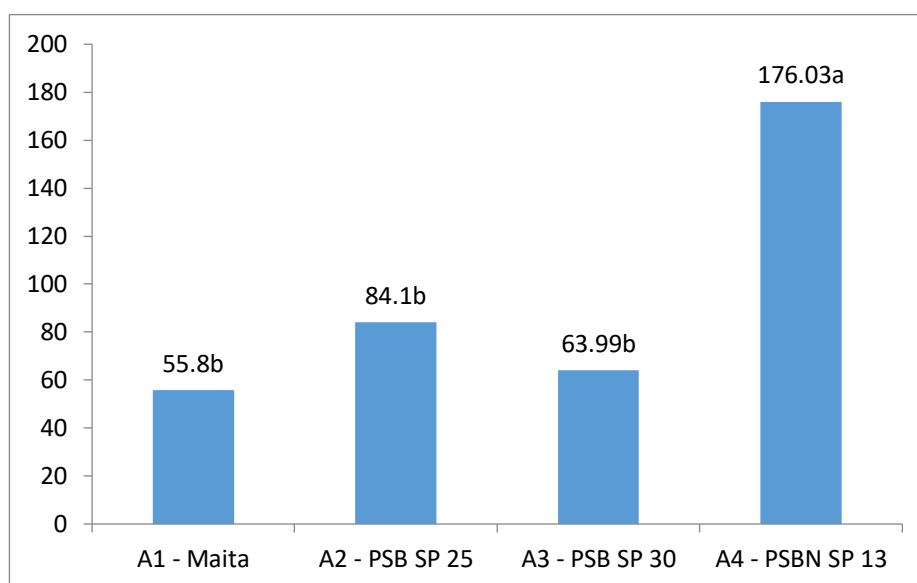


Figure 2. Number of nodes of sweet potato varieties as affected by the different CAPSU Formulated Soil Ameliorants (CFSA).

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	January 26, 2023
			Page	Page 9 of 20

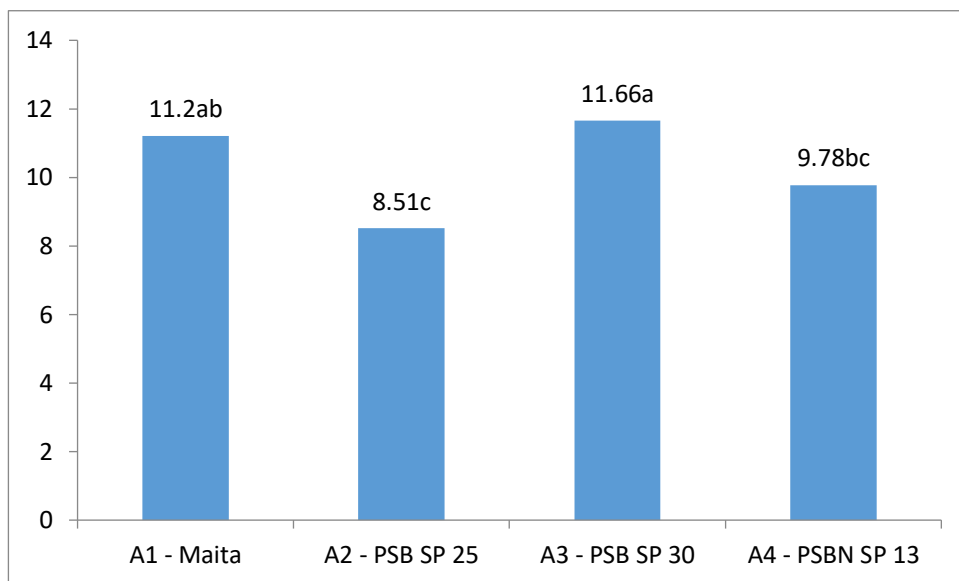


Figure 3. Length of leaves of sweet potato varieties as affected by the different CAPSU Formulated Soil Ameliorants (CFSA).

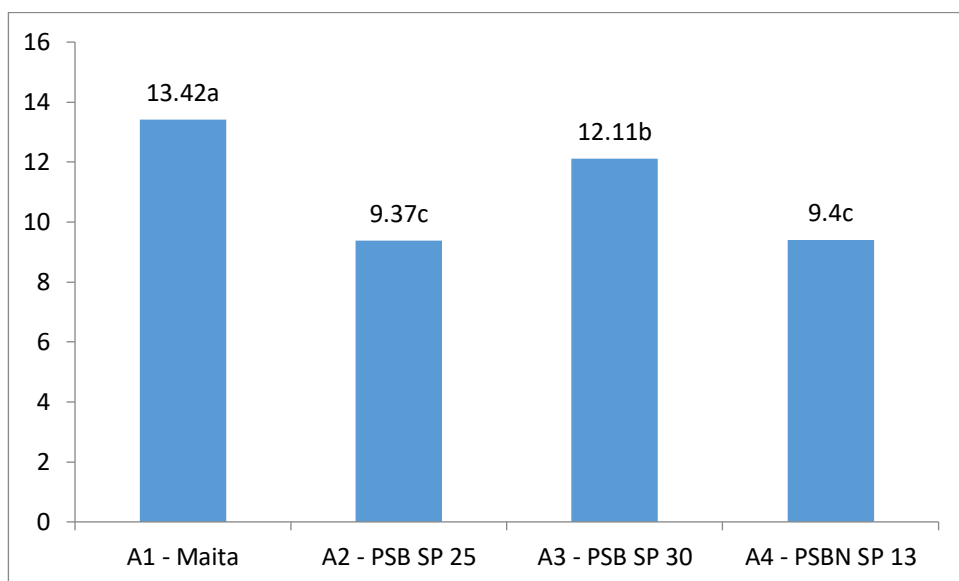


Figure 4. Width of leaves of sweet potato varieties as affected by the different CAPSU Formulated Soil Ameliorants (CFSA).

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	January 26, 2023
			Page	Page 10 of 20

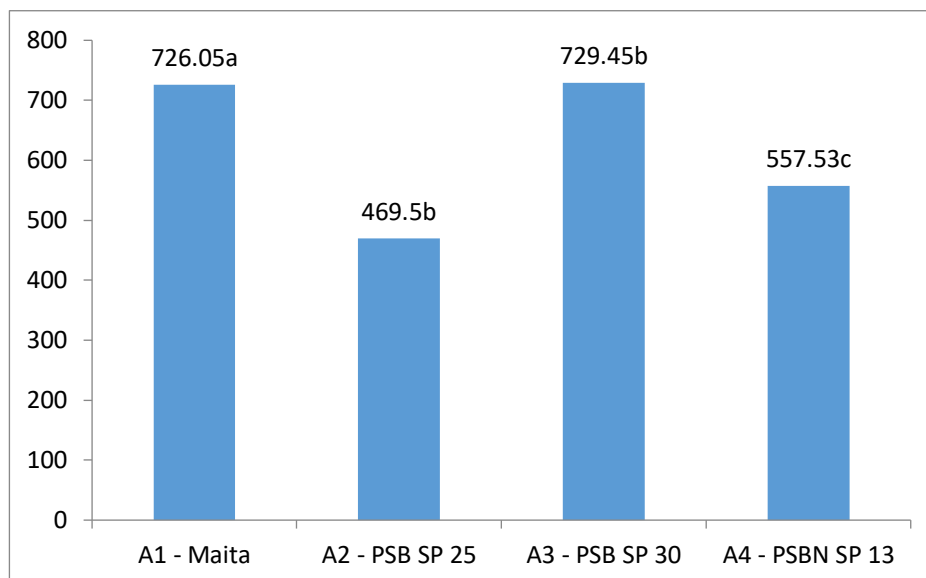


Figure 5. Biomass of sweet potato varieties as affected by the different CAPSU Formulated Soil Ameliorants (CFSA).

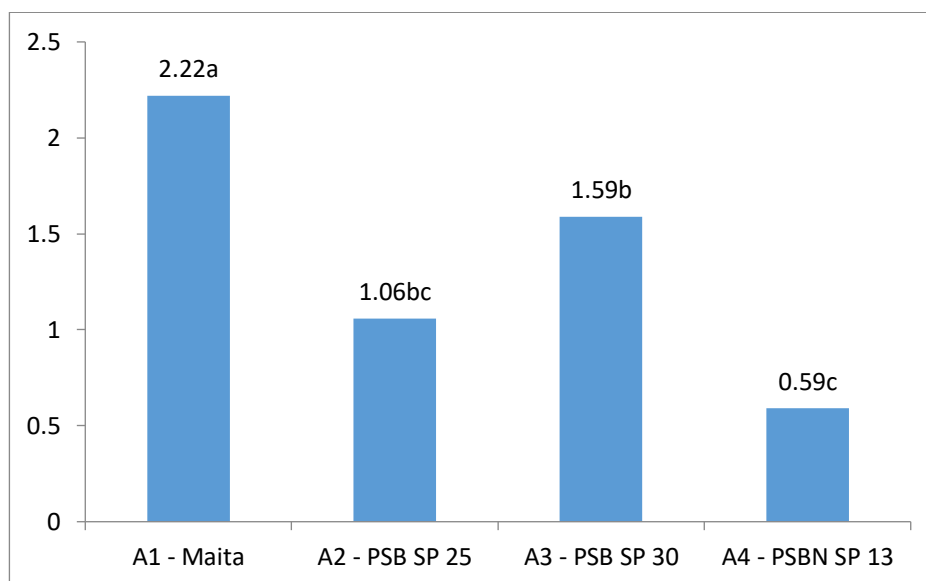


Figure 6. Number of tubers of sweet potato varieties as affected by the different CAPSU Formulated Soil Ameliorants (CFSA).

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	January 26, 2023
			Page	Page 11 of 20

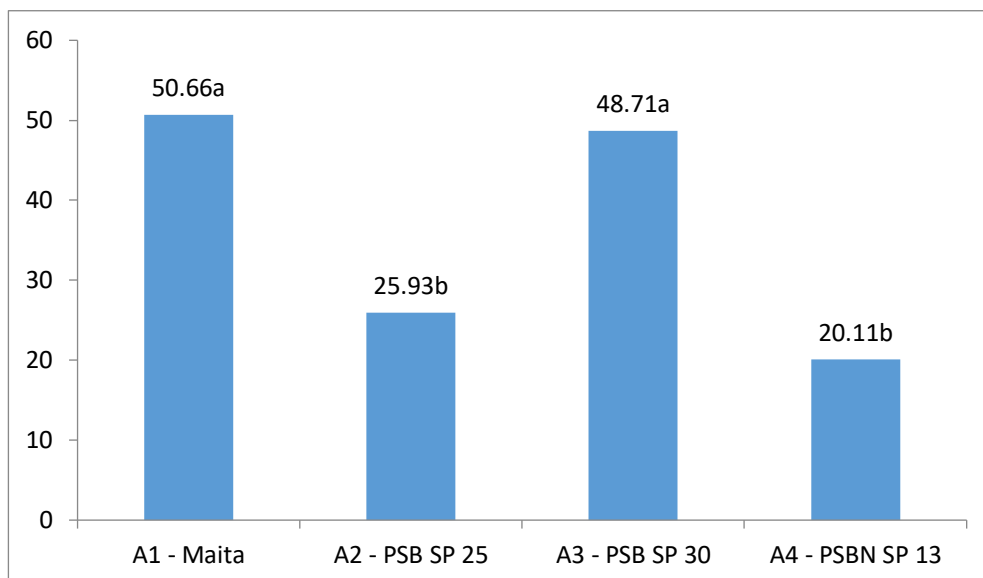


Figure 7. Diameter of tubers of sweet potato varieties as affected by the different CAPSU Formulated Soil Ameliorants (CFSA).

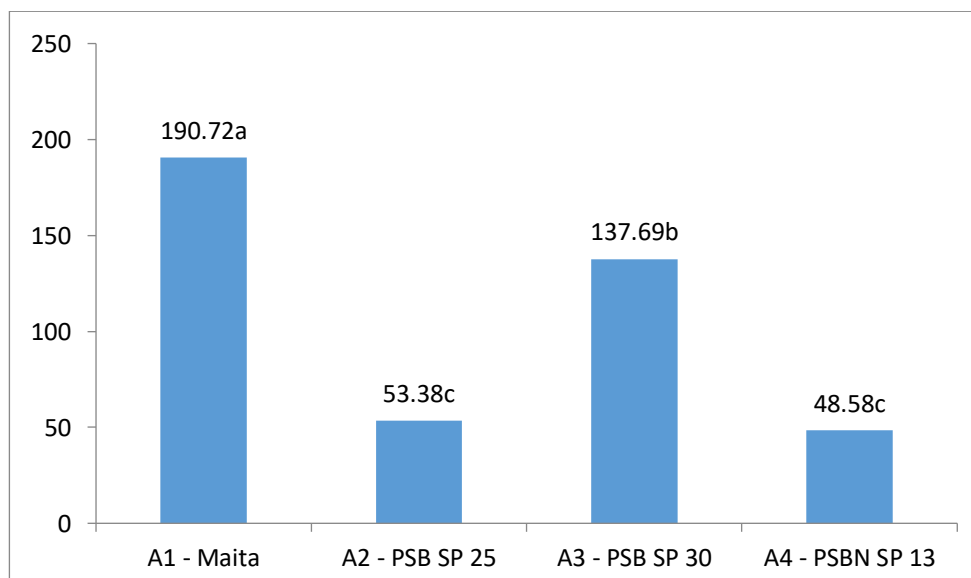


Figure 8. Weight of tubers of sweet potato varieties as affected by the different CAPSU Formulated Soil Ameliorants (CFSA).

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 12 of 20

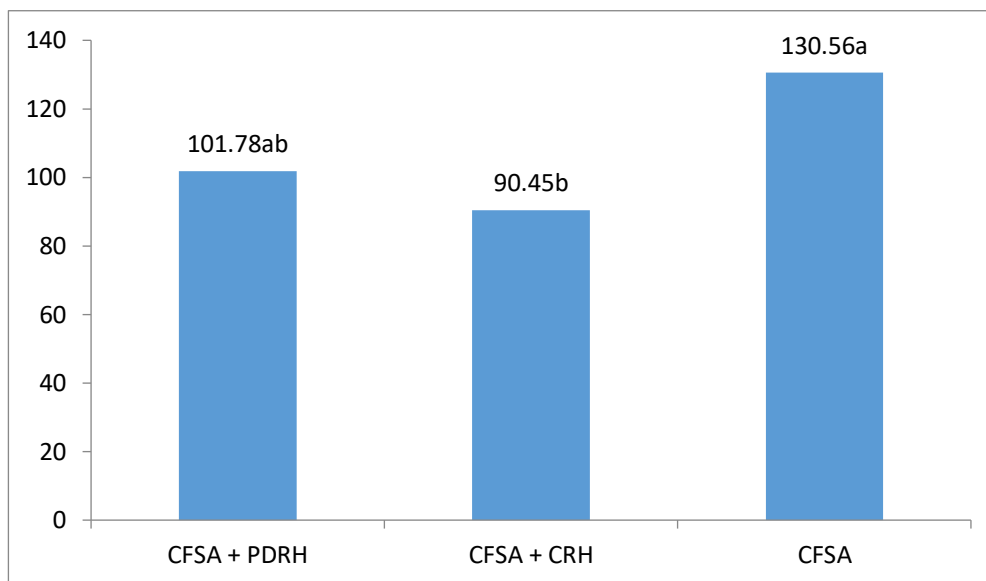


Figure 9. Weight of tubers of sweet potato varieties as affected by the different CAPSU Formulated Soil Ameliorants (CFSA).

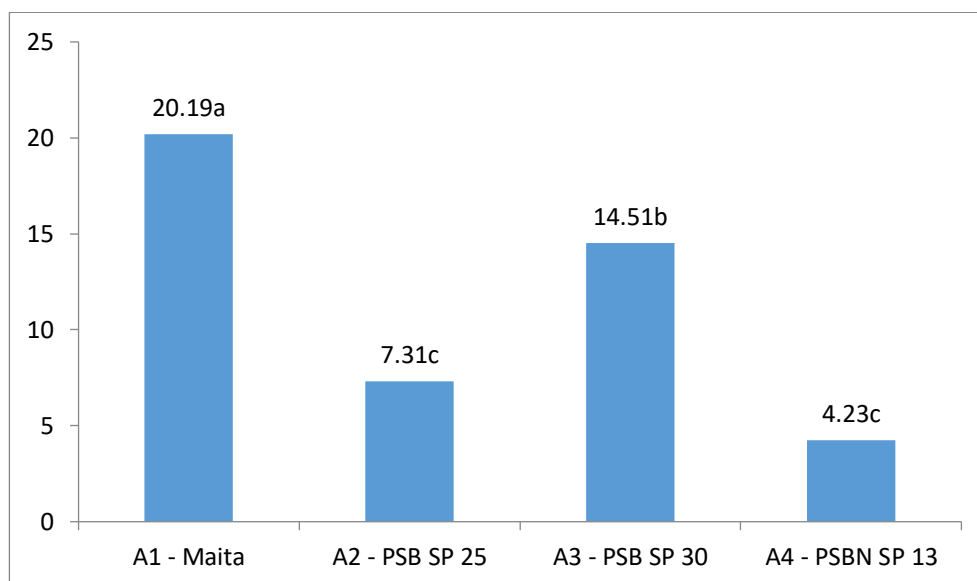


Figure 10. Length tubers of sweet potato varieties as affected by the different CAPSU Formulated Soil Ameliorants (CFSA).

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	January 26, 2023
			Page	Page 13 of 20

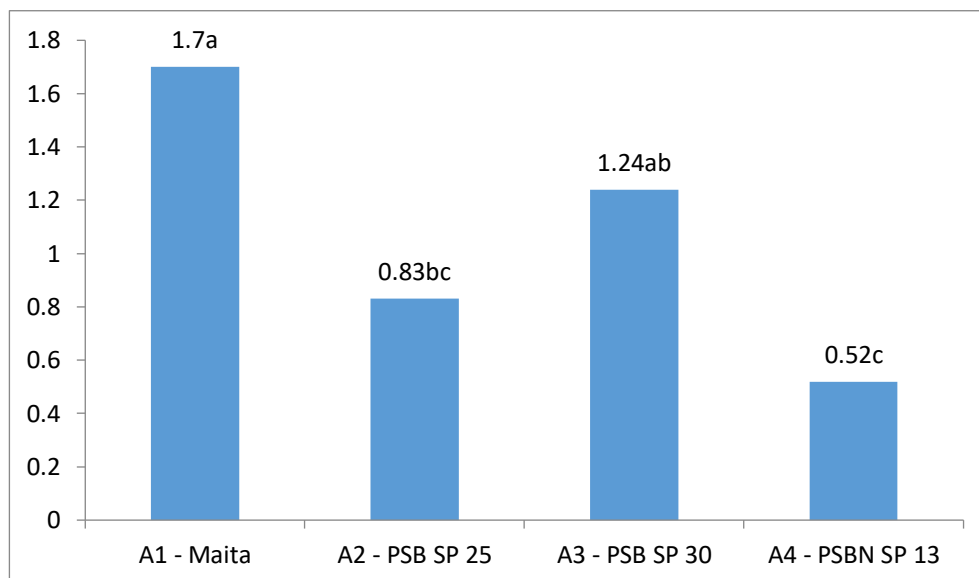


Figure 11. Number of marketable tuber of sweet potato varieties as affected by the different CAPSU Formulated Soil Ameliorants (CFSA).

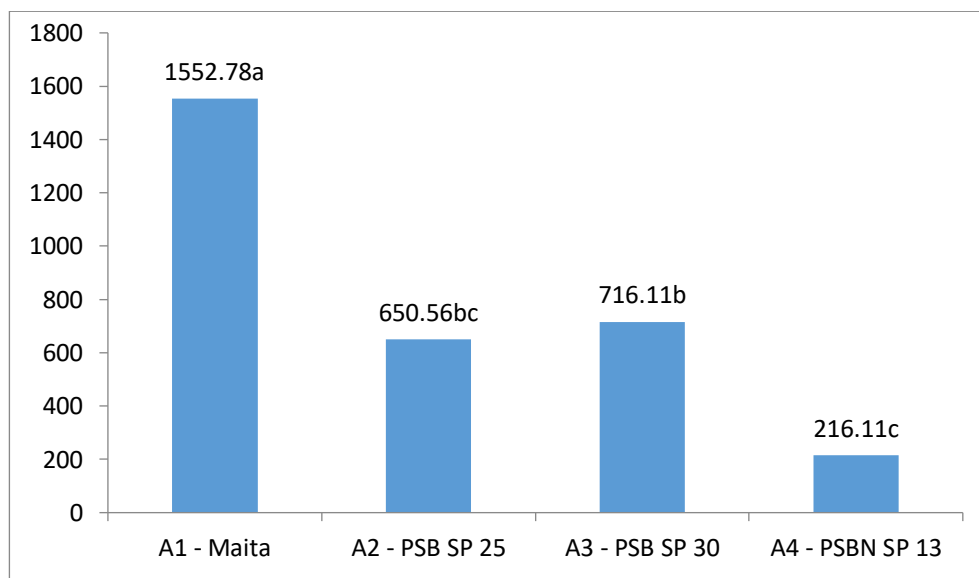


Figure 12. Yield per square meter of sweet potato varieties as affected by the different CAPSU Formulated Soil Ameliorants (CFSA).

	Document Type:	Document Code	RES-F08
	FORM ISO 9001:2015	Revision No.	00
	Document Title:	Effective Date	January 26, 2023
	COMPLETED RESEARCH	Page	Page 14 of 20

Table 1. Length of stem and stem girth of different sweet potato varieties applied with CAPSU Formulated Soil Ameliorants (CFSA).

Treatment	Length of Stem (cm)			Total	Mean	Treatment	Stem Girth ^{ns}			Total	Mean
	Soil Ameliorants						Soil Ameliorants				
Variety**	CFSA + PDRH	CFSA + CRH	CFSA			Variety	CFSA + PDRH	CFSA + CRH	CFSA		
A1 - Maita	271.36	262.88	269.77	804.01	268.00c	A1 - Maita	4.93	4.38	3.93	13.24	4.41
A2 - PSB SP 25	447.69	546.33	523.81	1517.83	505.94a	A2 - PSB SP 25	7.38	5	4.61	16.99	5.66
A3 - PSB SP 30	379.62	409.13	398.81	1187.56	395.85b	A3 - PSB SP 30	5.24	4.75	5.95	15.94	5.31
A4 - PSBN SP 13	537.26	648.68	453.44	1639.38	546.46a	A4 - PSBN SP 13	4.37	3.88	3.84	12.09	4.03
Total	1635.93	1867.02	1645.83			Total	21.92	18.01	18.33		
Mean	408.98	466.76	411.46			Mean	5.48	4.50	4.58		

	Document Type:	Document Code	RES-F08
	FORM ISO 9001:2015	Revision No.	00
	Document Title:	Effective Date	January 26, 2023
	COMPLETED RESEARCH	Page	Page 15 of 20

Table 2. Number of nodes and length of leaves of different sweet potato varieties applied with CAPSU Formulated Soil Ameliorants (CFSA).

Treatment	Number of nodes					Treatment	Length of Leaves				
	Soil Ameliorants						Soil Ameliorants				
Variety*	CFSA + PDRH	CFSA + CRH	CFSA	Total	Mean	Variety**	CFSA + PDRH	CFSA + CRH	CFSA	Total	Mean
A1 - Maita	57.17	55.3	54.93	167.4	55.80b	A1 - Maita	11.511	10.39	11.69	33.591	11.20ab
A2 - PSB SP 25	76.53	91.6	84.17	252.3	84.10b	A2 - PSB SP 25	8.37	8.87	8.28	25.52	8.51c
A3 - PSB SP 30	60.4	63.73	67.83	191.96	63.99b	A3 - PSB SP 30	10.56	10.76	13.67	34.99	11.66a
A4 - PSBN SP 13	89.53	230.73	207.84	528.1	176.03a	A4 - PSBN SP 13	10.07	9.65	9.63	29.35	9.78bc
Total	283.63	441.36	414.77			Total	40.511	39.67	43.27		
Mean	70.91	110.34	103.69			Mean	10.13	9.92	10.82		

	Document Type:	Document Code	RES-F08
	FORM ISO 9001:2015	Revision No.	00
	Document Title:	Effective Date	January 26, 2023
	COMPLETED RESEARCH	Page	Page 16 of 20

Table 3. Width of leaves and biomass of different sweet potato varieties applied with CAPSU Formulated Soil Ameliorants (CFSA).

Treatment	Width of Leaves			Total	Mean	Treatment	Biomass			Total	Mean
	Soil Ameliorants						Soil Ameliorants				
Variety**	CFSA + PDRH	CFSA + CRH	CFSA			Variety**	CFSA + PDRH	CFSA + CRH	CFSA		
A1 - Maita	13.1	14.03	13.12	40.25	13.42a	A1 - Maita	716.5	620.77	840.87	2178.14	726.05a
A2 - PSB SP 25	9.54	9.15	9.43	28.12	9.37c	A2 - PSB SP 25	489.73	487.27	431.5	1408.5	469.50b
A3 - PSB SP 30	12.05	11.93	12.34	36.32	12.11b	A3 - PSB SP 30	683.5	743.17	761.67	2188.34	729.45a
A4 - PSBN SP 13	9.81	9.47	8.93	28.21	9.40c	A4 - PSBN SP 13	635.17	509.83	527.6	1672.6	557.53b
Total	44.5	44.58	43.82			Total	2524.9	2361.04	2561.64		
Mean	11.13	11.15	10.96			Mean	631.23	590.26	640.41		

	Document Type:	Document Code	RES-F08
	FORM ISO 9001:2015	Revision No.	00
	Document Title:	Effective Date	January 26, 2023
	COMPLETED RESEARCH	Page	Page 17 of 20

Table 4. Number and diameter of tubers of different sweet potato varieties applied with CAPSU Formulated Soil Ameliorants (CFSA).

Treatment	Number of tubers			Total	Mean	Treatment	Diameter of tubers			Total	Mean
	Soil Ameliorants						Soil Ameliorants				
Variety**	CFSA + PDRH	CFSA + CRH	CFSA			Variety*	CFSA + PDRH	CFSA + CRH	CFSA		
A1 - Maita	2.67	1.93	2.07	6.67	2.22a	A1 - Maita	54.22	46.75	51	151.97	50.66a
A2 - PSB SP 25	1.3	0.9	0.97	3.17	1.06bc	A2 - PSB SP 25	25.09	24.98	27.73	77.8	25.93b
A3 - PSB SP 30	1.59	1.82	1.35	4.76	1.59b	A3 - PSB SP 30	61.28	25.63	59.21	146.12	48.71a
A4 - PSBN SP 13	0.5	0.93	0.33	1.76	0.59c	A4 - PSBN SP 13	16.01	28.09	16.23	60.33	20.11b
Total	6.06	5.58	4.72			Total	156.6	125.45	154.17		
Mean	1.52	1.40	1.18			Mean	39.15	31.36	38.54		

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	January 26, 2023
			Page	Page 18 of 20

Table 5. Weight and length tubers of different sweet potato varieties applied with CAPSU Formulated Soil Ameliorants (CFSA).

Treatment	Weight of tubers			Total	Mean	Treatment	Length of tubers			Total	Mean
	Soil Ameliorants*						Soil Ameliorants				
Variety**	CFSA + PDRH	CFSA + CRH	CFSA			Variety**	CFSA + PDRH	CFSA + CRH	CFSA		
A1 - Maita	190.6	144.57	237	572.17	190.72a	A1 - Maita	17.85	26.15	16.57	60.57	20.19a
A2 - PSB SP 25	46.61	49.67	63.87	160.15	53.38c	A2 - PSB SP 25	6.72	7.16	8.04	21.92	7.31c
A3 - PSB SP 30	123.83	111.55	177.7	413.08	137.69b	A3 - PSB SP 30	14.93	13.84	14.77	43.54	14.51b
A4 - PSBN SP 13	46.07	56	43.68	145.75	48.58c	A4 - PSBN SP 13	4.78	5.54	2.38	12.7	4.23c
Total	407.11	361.79	522.25			Total	44.28	52.69	41.76		
Mean	101.78ab	90.45b	130.56a			Mean	11.07	13.17	10.44		

	Document Type:	Document Code	RES-F08
	FORM ISO 9001:2015	Revision No.	00
	Document Title:	Effective Date	January 26, 2023
	COMPLETED RESEARCH	Page	Page 19 of 20

Table 6. Number of marketable and non-marketable tuber of different sweet potato varieties applied with CAPSU Formulated Soil Ameliorants (CFSA).

Treatment	Number of marketable tuber			Total	Mean	Treatment	Number of non-marketable tuber			Total	Mean
	Soil Ameliorants						Soil Ameliorants				
Variety**	CFSA + PDRH	CFSA + CRH	CFSA			Variety**	CFSA + PDRH	CFSA + CRH	CFSA		
A1 - Maita	1.83	1.57	1.7	5.1	1.70a	A1 - Maita	1.10	0.96	0.92	2.98	0.99
A2 - PSB SP 25	1.07	0.7	0.73	2.5	0.83bc	A2 - PSB SP 25	0.70	1	0.83	2.53	0.84
A3 - PSB SP 30	1.1	1.1	1.53	3.73	1.24ab	A3 - PSB SP 30	0.60	0.33	0.33	1.26	0.42
A4 - PSBN SP 13	0.5	0.72	0.33	1.55	0.52c	A4 - PSBN SP 13	0	0.67	0	0.67	0.22
Total	4.5	4.09	4.29			Total	2.40	2.96	2.08		
Mean	1.13	1.02	1.07			Mean	0.60	0.74	0.52		

	Document Type:	Document Code	RES-F08
	FORM ISO 9001:2015	Revision No.	00
	Document Title:	Effective Date	January 26, 2023
	COMPLETED RESEARCH	Page	Page 20 of 20

Table 7. Yield per square meter of different sweet potato varieties applied with CAPSU Formulated Soil Ameliorants (CFSA).

Treatment	Yield per Square Meter				
	Soil Ameliorants			Total	Mean
Variety**	CFSA + PDRH	CFSA + CRH	CFSA		
A1 - Maita	1535	1478.33	1645	4658.33	1552.78a
A2 - PSB SP 25	636.67	643.33	671.67	1951.67	650.56bc
A3 - PSB SP 30	830	631.67	686.67	2148.34	716.11b
A4 - PSBN SP 13	196.67	246.67	205	648.34	216.11c
Total	3198.34	3000	3208.34		
Mean	799.59	750.00	802.09		